

HP-WAN Progress and Documents Update

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HP-WAN IETF125 Hackathon Progress

- **Where we are:**

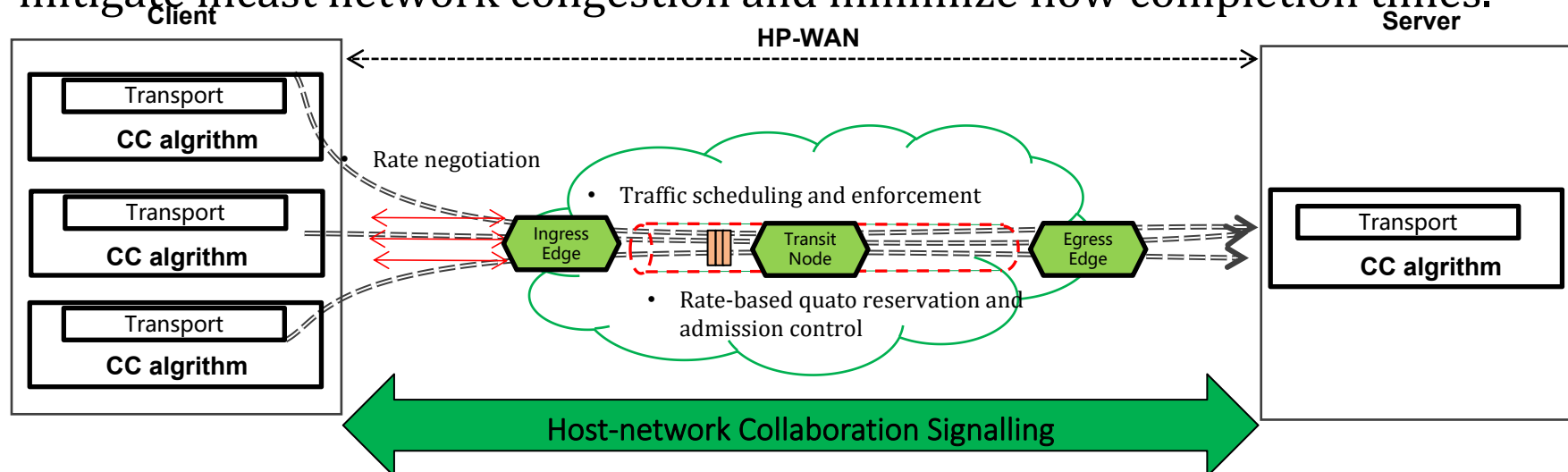
- Check HPWAN state of art for different transport domain technologies.
- Identify methods for end-to-end HPWAN service monitoring and which IETF performance monitoring and tooling is suitable.
- Discuss HPWAN deployment on topologies for public networks and service scenarios based on the HP-WAN framework.
- Test the integration and simulation of the HP-WAN functions, such as rate negotiation, admission control, traffic scheduling, and resource reservations with distributed signalling (IETF RSVP-TE).
- Test the RDMA performance and QUIC-based optimization simulation.

Two Important Documents Update

- **“Current State of the Art for High Performance Wide Area Networks”**
 - This document provides an overview of the terminology and techniques used for existing HP-WANs. It also explores the technological advancements, operational tools, and future directions for HP-WANs, emphasising their role in enabling cutting-edge scientific research, AI training and massive R&E data analysis.
 - <https://datatracker.ietf.org/doc/draft-kcrh-hpwan-state-of-art/>
 - Next step: request to ISE AD sponser
- **Framework for High Performance Wide Area Network (HP-WAN)**
 - This document defines a framework to enable the host-network collaboration for high-speed and high-throughput data transmission, coupled with fast completion time and newtork utilization of HP-WAN. It can be realized via either distributed host-network collaborative signaling or centralized configuration.
 - <https://datatracker.ietf.org/doc/draft-xhy-hpwan-framework/>
 - Update: modify the traffic workflow to proactively enforce rate control from network side and mitigate pre-congestion conditions
 - Next step: revision and present in TSVWG

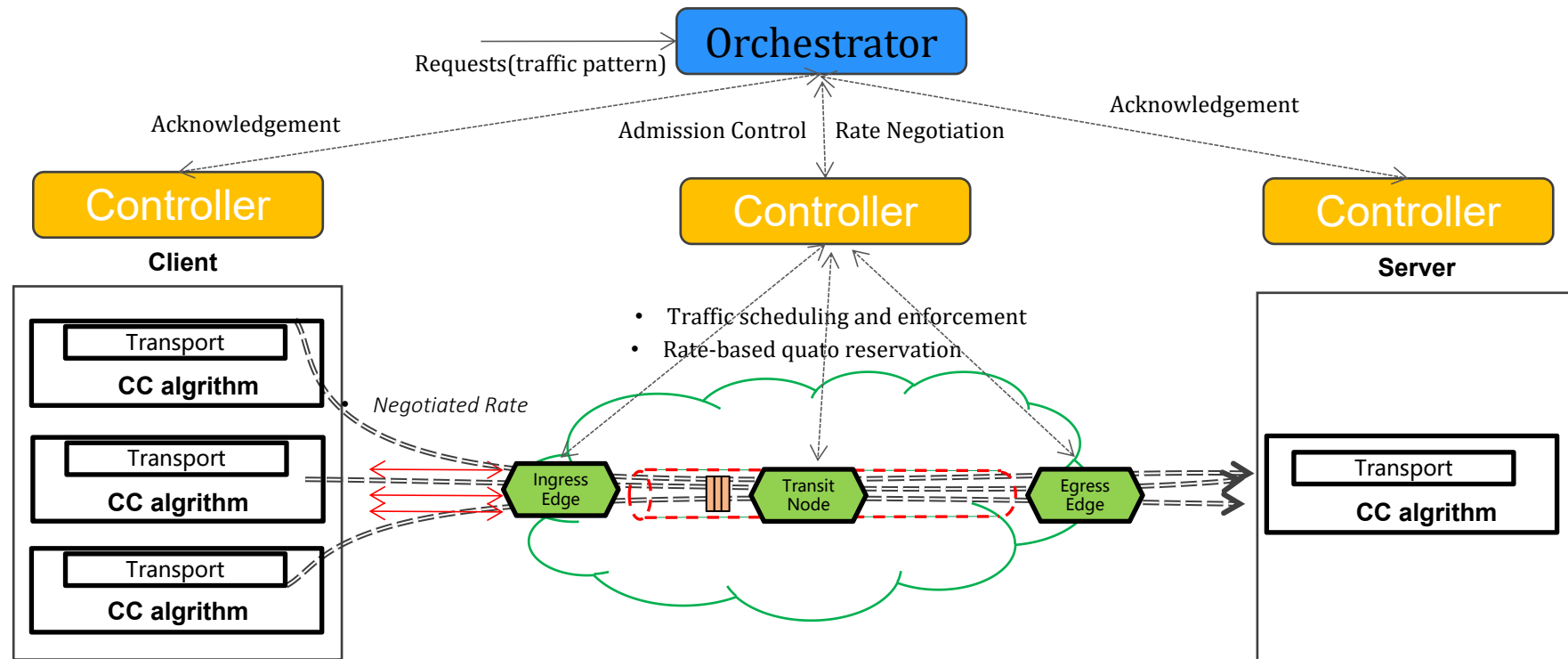
Host-network Collaboration: Distributed Signaling

- The host-network **collaboration signaling** is enabled to enhance the **congestion control** in HP-WAN.
 - The client and server could send the traffic efficiently with the negotiated rate-based congestion control in a fine-grained way.
 - The network could provide rate-based quota reservation and admission control to achieve predictable network behaviour.
 - The edge node could enhance the capability to provide traffic scheduling and enforcement to mitigate incast network congestion and minimize flow completion times.



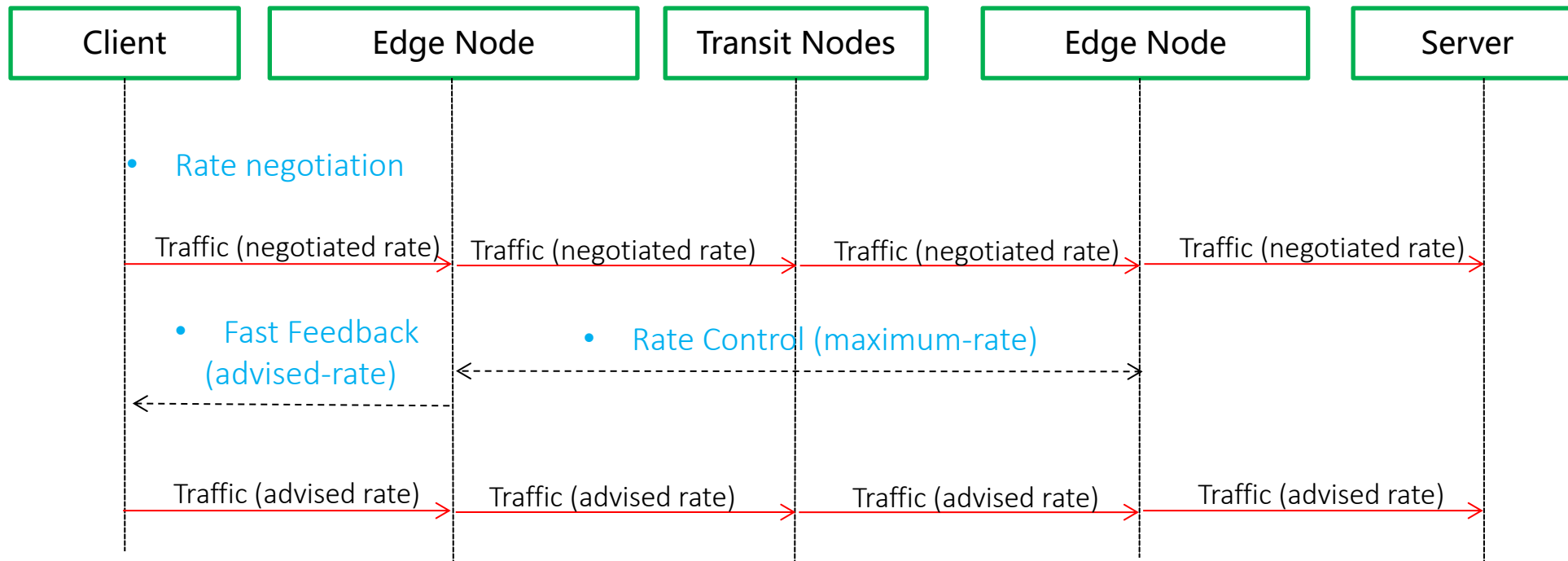
Host-network Collaboration: Centralized Configuration

- Host-and-network collaboration could also be performed using configuration in centralized model where **a controller or an orchestrator orchestrates data processing and resource allocation** across hosts and network infrastructure.
 - The request of scheduled traffic will be initiated to the Orchestrator.
 - The Orchestrator will perform rate negotiation among hosts and networks.
 - The controller of the network will perform admission control and acknowledge the traffic.
 - The Acknowledgement will be configured to the client, including the response with the negotiated rate and QoS policy for the client to send traffic.
 - The Controller of the network will reserve resource quota to guarantee the job's completion time.



Host-network Collaboration: Traffic Workflow

- Host-and-network collaboration could also be performed within the **workflow of traffic transmission**.
 - The client could send traffic according to the negotiated rate policy to achieve a high throughput within the completion time.
 - The edge node could adjust the maximum rate of traffic when the network performs the rate control in pre-congestion situation.
 - The edge node also could send fast feedback with the advised rate when the traffic rate does not apply to the network..



- **Thanks !**
- **Comments and suggestions are welcome.**